

PROJECT SYNOPSIS

MudraTrack — Advanced GST Invoicing & Job-Card ERP

Enterprise ERP for Print Studios, Banner Manufacturers & Digital Packaging

STUDENT NAME

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ORGANIZATION

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01 PROJECT OVERVIEW

MudraTrack is an enterprise-grade ERP and billing application built to streamline operations across print shops, digital packaging studios, and banner manufacturers. The system bridges the gap between job card production pipelines and a compliant GST / E-Invoicing billing system.

It supports role-based access control (RBAC), a prepaid company wallet recharged via Razorpay, automatic notification systems (WhatsApp & OAuth Email), and an AI-powered conversational agent utilizing Google Gemini — allowing executives to run complex, natural-language business reports and audits on their sales ledger securely.

02 TECHNOLOGIES USED

Frontend	Backend
• React.js (v18) — Functional Components & Hooks • Redux Toolkit + Redux Persist (State Management) • Apollo Client — GraphQL Query / Mutation / Cache • Material UI (MUI) — Enterprise UI Components • Framer Motion — Animations & Transitions	• NestJS + TypeScript — Scalable API Server • GraphQL + Apollo Server — Schema-First API • Prisma ORM — Type-Safe DB Access • MongoDB — NoSQL Document Database • Node-cron — Scheduled Jobs / Recurring Invoices
Cloud Infrastructure	Third-Party Integrations
• AWS Cognito — Authentication & Session Tokens • AWS S3 — Invoice PDF & Image Storage • AWS SES / SNS — Email & SMS Notifications • Nx Monorepo CLI — Build & Dependency Management	• Razorpay Payment Gateway — Wallet Recharge • Google Gemini LLM — AI Business Intelligence • WhatsApp Business API (WABA) — Notifications • OAuth 2.0 — Gmail / Microsoft Outlook Integration

03 WORK DONE BY ME

FRONTEND ENHANCEMENTS

Multi-Company Switching Workflow

Engineered a seamless tenant/company switching mechanism on the dashboard with automatic state synchronization using Apollo Cache — allowing users to switch companies without full page reload.

■ **Mobile Responsiveness & UI Refactoring**

Refactored core dashboard pages, navigation, and input forms (payment recharge forms, company profile editors) to ensure clean cross-device UI rendering across all screen sizes.

■ **Multi-Language Localization Support**

Integrated Marathi translation support and repaired Hindi localization files (hi.json and mr.json) with proper character encoding for regional invoicing and dashboard experiences. Full project localization across all modules.

■ **Payment Recharge Interface (WalletTab)**

Built the frontend layout for WalletTab and integrated the official Razorpay Checkout SDK to handle secure online wallet recharges with real-time payment status feedback.

■ **Bug Fixes & Code Quality**

Identified and resolved multiple UI/UX bugs across the dashboard — including state inconsistencies, form validation edge cases, and rendering issues on low-resolution screens.

BACKEND DEVELOPMENT

■ **Quantity-Wise Dynamic Pricing Engine**

Programmed the dynamic pricing calculator that automatically applies volume-based slab discounts and calculates item totals on invoices based on quantity limits — supporting customer-specific override rates.

■ **Razorpay Payment Gateway Integration**

Designed backend routes for creating Razorpay payment orders, validating cryptographic webhook signatures (x-razorpay-signature via SHA256 HMAC), and automated wallet crediting via atomic Prisma database transactions.

■ **WhatsApp Notification System**

Implemented a WhatsApp transactional message dispatcher using WhatsApp Business API (WABA) to send instant order confirmations and digital receipt links to clients automatically.

■ **Database Schema Design**

Extended and optimized MongoDB collection structures for WalletRechargeTransaction and double-entry transaction ledgers using Prisma ORM schemas with full type safety.

04 KEY OUTCOMES & LEARNINGS

Full-Stack ERP Development

Built production-ready features across React frontend and NestJS backend in a real enterprise environment.

Localization & i18n

Successfully localized entire application in Hindi and Marathi with proper Unicode encoding.

Database Design

Designed scalable MongoDB schemas with Prisma ORM for financial transaction ledgers.

Payment Integration

Implemented secure Razorpay payment gateway with cryptographic webhook verification.

AI Integration

Gained hands-on experience integrating Google Gemini LLM with function-calling for business intelligence.

Team Collaboration

Worked in a professional Nx monorepo environment with CI/CD pipelines and code review workflows.